CHAPTER I

INTRODUCTION

INTRODUCTION

1.1 PROBLEM DEFINITION

During such emergency incidents as fires, industrial accidents, and the search and rescue process, visibility is typically severely reduced by smoke, haze, and additional environmental obstacles. That poor visibility limits the ability of first responders to assess conditions, locate people in danger and traverse risky terrain, ultimately putting both victims and emergency personnel at greater risk. The repercussions of inadequate visibility leading to delayed decision-making can create life-threatening environment and make the effective rescue difficult.

Existing approaches for visibility augmentation, such as thermal cameras and classical image processing, are its simple solutions. Thermal cameras can detect hotspots, but lack detailed scene information in general, so it is impossible to separate objects and humans. Although popular, traditional dehazing algorithms, which are typically utilized in post-processing, are too slow for real time execution and therefore irrelevant in instantaneous emergency situations where immediate visibility is desired.

To solve these problems, it is suggested that an already existing real-time, drone-based dehazing system be put in place to support visibility in areas impacted by smoke while simultaneously detecting the presence of humans. This allows responders equipped with airborne drones that carry high-performance dehazing algorithms and A.I. human detection systems to have an improved view of the scene, improving situational awareness, decreasing search time, and enabling faster, safer and more effective search and rescue missions.

1.2 PURPOSE OF STUDY

The main context area of the research done on this thesis is centered around the (not very novel) development and utility validation of a drone-based dehazing system which is expected to increase visibility in densely smoked areas, enabling first responders to carry out their missions more efficiently. By minimizing the effects of smoke and haze, the system will provide optimized real-time video enhancement so that firefighters, search-and-rescue teams, and other emergency responders can more effectively operate in life-threatening situations.

Second, the objective of the research is to design the system, which can detect the humans and automatically mark the people who are in the distress. And, such a feature plays a crucial role

to increase efficiency in rescue missions by decreasing search time and providing a quicker response to people in need.

This research hopes to explore and investigate these aspects through the creation and development of a dehazing application with drones.

i. Enhance Visibility – Deploy and confirm dehazing algorithms capable of generating clear, high–fidelity imagery in real time within a smoke serviceable environment.
ii. Increase Situational Awareness – Provide emergency responders with access to a rapid, accurate indicator to assess quickly and optimise decisions.
iii. Maximize Search and Rescue – Identify subjects faster, even in low visibility, with AI human detection.
iv. Real-time Performance – Build a system that operates with little latency and always stable visual clarity in fast-paced emergencies.
v. Validate System Effectiveness – Perform testing in simulated environments, quantifying accuracy and reliability of both the dehazing module and the human detector module.

An in-context, drone-based solution to visualization challenges, this research contributes to the body of emergency response technology and research around technology that improves responder and victim safety outcomes.

1.3 RESEARCH QUESTIONS

This study aims to explore the following key research areas:

- To what extent can the proposed drone system enhance visibility in smoke-obscured environments using real-time dehazing algorithms?
- How dependable and precise is the integrated human detection capability under low-visibility conditions?
- In what ways does utilizing drone-mounted dehazing technology improve situational awareness for emergency teams compared to conventional visibility enhancement methods?
- What specific limitations and challenges emerge when deploying real-time dehazing and human detection on drones during emergency operations?

- How does system latency affect the usability and operational effectiveness of the system in live emergency response scenarios?

1.4 SIGNIFICANCE OF STUDY

The study has several implications and its importantance:

i. Improved Emergency Response Efficiency – Since it gives better visibility in smoke conditions, the system can assist firefighters, search-and-rescue units, and other emergency responders to react more quickly and efficiently. This can decrease rescue times and fatalities.

ii. Enhanced Safety for Responders and Victims – Poor visibility is a critical risk to victims and emergency responders. The suggested system reduces such risks by ensuring real-time clarity, enabling responders to move through dangerous environments safely.

iii. Technological Progress in Dehazing – Although there are conventional dehazing methods, most are computationally costly and not real-time optimized.This work makes an addition to the development of effective, quick, and AI-driven dehazing algorithms optimized for emergency response.

iv. Integration of AI for Human Detection – The combination of dehazing and human detection renders this system a double solution that enhances visibility as well as facilitates identification of individuals in distress. Integration is particularly justifiable for search-and-rescue missions.

v. Real-World Applicability – The outcomes of this study are applicable to be adopted by emergency response bodies and incorporated into existing drone-rescue operations. The technology also holds promise for use in beyond firefighting emergencies such as industrial accidents, military combat, and disaster response.

vi. Ground for Future Studies – This study lays a ground for future developments in drone-based emergency response systems. Future research can improve upon this to improve AI algorithms, increase hardware efficiency, and apply the technology in other visibility-obstructed situations.

1.5 OVERVIEW

This project aims to create a drone-based dehazing system to improve visibility in a smoke-filled environment, enabling emergency responders to make faster, safer decisions. By combining real-time dehazing functionality with AI-based human detection, the system offers better views and detects people in need of rescue, maximizing search-and-rescue missions. Unlike traditional methods, this solution supports low-latency functionality and hence is of tremendous value to dynamic emergency situations. The research contributes to the advancement of emergency response technology, promoting situational awareness, and developing the building blocks for future advancements in AI-driven aerial surveillance.

CHAPTER II

LIETERATURE REVIEW

LITERATURE REVIEW

2.1. KEY RESEARCH CHALLENGES

Developing an emergency response-compatible real-time, drone-mounted dehazing system presents several challenges that must be addressed. The dominant challenge is real-time processing since dehazing and human detection are computationally intensive AI-based models, although drones possess limited onboard processing power. Processing with low latency and little dependence on clouds is required. In addition, density of smoke and air conditions vary based on parameters like fire strength and wind, and thus it is difficult to create a dehazing model that can be used for all cases.

The second most important challenge is human detection in low-visibility environments, as thick smoke causes occlusions and reduces contrast, making it difficult to distinguish people from rubble. Furthermore, drones are power and flight-time constrained because AI-based processing increases battery consumption, reducing operational time. Quality video streaming in real-time is also critical, but network interferences, smoke bias, and damaged infrastructure can distort streaming quality. Secondly, the lack of on-ground data for training AI models impedes robust solution formulation that is practical in dynamic emergency settings. Lastly, regulatory, privacy, and ethical concerns of drone deployment over public spaces must be addressed to ensure compliance with aviation laws as well as compliance with data privacy. This can be best obtained through a trade-off between AI effectiveness, hardware efficiency, and real-world scalability to optimize situational awareness during emergency response.

2.2 EXISTING RESEARCH AND DEVELOPMENTS

Table 2.1 Literature Survey

Sr. No	Title of Paper	Year of Publication	Publisher	Methodology	Conclusion
1.	A Fire and Smoke Detection Model Based on YOLOv8 Improvement.	2024	IEEE	The authors proposed using GradCAM and YOLO to identify fire intensity and temperature.	The authors claim this method is effective for fire detection but

					insufficient for handling smoke-induced visibility issues.
2.	Object Detection through Fires Using Violet Illumination Coupled with Deep Learning.	2023	MDPI	The authors proposed a technique combining violet illumination with YOLOv5 to enhance object detection through flames.	The authors claim this approach effectively mitigates flame obstructions but does not significantly improve visibility in dense smoke.
3.	Multi-stages de-smoking model based on CycleGAN for surgical de-smoking.	2023	Springer	The authors proposed a multi-stage CycleGAN model for surgical smoke removal, focusing on color and contrast preservation.	The authors claim the model performs well in medical settings but is not applicable to dynamic emergency scenarios.
4.	Self-Supervised Video Desmoking for Laparoscopic Surgery	2024	arXiv	The authors proposed a self-supervised video desmoking method tailored for light	The authors claim it is effective in limited conditions but

				smoke in controlled environments.	not robust for real-time emergency use.
5.	An Algorithm of Battlefield Image Desmoking Based on Semantic Guidance and Contrastive Learning.	2024	Acta Armamentarii	The authors proposed SCLGAN, a semantic-guided contrastive learning algorithm for efficient desmoking.	The authors claim it outperforms previous methods but is focused on battlefield imagery, not rescue operations.
6.	Single image desmoking via attentive generative adversarial network for smoke detection process.	2021	Springer	The authors proposed an attentive GAN model that integrates attention mechanisms to reduce smoke interference.	The authors claim it enhances image quality, but its high computational cost limits real-time application.
7.	FD-GAN: Generative Adversarial Networks with Fusion-Discriminator for Single Image Dehazing.	2020	AAAI	The authors proposed FD-GAN, which applies frequency-based enhancements for smoke removal.	The authors claim the method achieves effective dehazing but requires high computational resources.

8.	TMS-GAN: A Twofold Multi-Scale Generative Adversarial Network for Single Image Dehazing	2022	IEEE	The authors proposed a multi-scale GAN (TMS-GAN) to improve dehazing across varying image scales.	The authors claim it enhances image clarity but is unsuitable for UAV-based real-time applications.
9.	Lightweight Neural Network for the Real-Time Dehazing of Tidal Flat UAV Images Using a Contrastive Learning Strategy. Drones.	2024	MDPI	The authors proposed a lightweight neural network optimized for drone-based dehazing with minimal latency.	The authors claim it achieves 0.523ms latency but lacks adaptation to dense smoke or human detection.
10.	UCL-Dehaze: Toward Real-World Image Dehazing via Unsupervised Contrastive Learning.	2024	IEEE	The authors proposed UCL-Dehaze, an unsupervised contrastive learning model for real-world drone dehazing.	The authors claim it delivers smooth performance with 80ms runtime but does not address emergency-specific challenges.

The problem of enhancing visibility in smoky environments has been solved by different methods, such as deep learning, image processing, and hardware-based solutions. Some research has concentrated on fire and smoke detection, employing AI models like GradCAM and YOLO for fire intensity and temperature identification (IEEE, 2024). While these techniques work well in indicating the existence of fire, they are not enough in addressing the issue of visibility provided by thick smoke. Another study investigated violet lighting with YOLOv5 for enhancing object detection by flames (MDPI, 2023); however, this method primarily identifies fire obstructions and not smoke, which remains the predominant issue in hindering visibility.

For dehazing and desmoking, several deep learning methods have been suggested. Multi-Stage CycleGAN model (Springer, 2023) has also been found to perform for surgical de-smoking with improved color and contrast artifact retention compared to conventional techniques. Likewise, self-supervised desmoking of video through SelfSVD has been found to perform even with faint smokes under lab conditions (arXiv, 2024). Yet, both techniques are tailored to medical applications as opposed to dynamic emergency situations and hence are not applicable to actual-time disaster response. Another interesting study proposed SCLGAN, a semantic-guided contrastive learning framework, which performed better than earlier desmoking methods (Acta Armamentarii, 2024). Although it is effective, its primary intention is image enhancement in the battlefield as opposed to emergency rescue operations.

Current advancements in Generative Adversarial Networks (GANs) have significantly contributed to image dehazing. Studies such as Single Image Desmoking via Attentive GAN (Springer, 2021) and FD-GAN (AAAI, 2020) employ attention mechanisms and frequency-based enhancements to reduce smoke interference. TMS-GAN (IEEE, 2022) also proposes a multi-scale method for better real-world dehazing. Although these methods improve image quality, they tend to have high computational requirements, rendering them inappropriate for real-time processing on resource-constrained UAV platforms.

For dehazing using drones, a lightweight neural network (MDPI, 2024) was proposed for real-time dehazing of UAV images with a latency of 0.523ms. Similarly, UCL-Dehaze (IEEE, 2024) came up with an unsupervised contrastive learning framework, which introduced smooth real-world dehazing with a mean runtime of 80ms. These research papers show progress toward real-time UAV dehazing, but none of them are aimed at dense smoke scenarios relevant to emergency response missions.

While there are current endeavors towards fire detection, image dehazing, and visibility enhancement with UAVs, they lack an integrated approach to both real-time smoke elimination and human detection. Current studies either create current solutions towards fire detection without dehazing, dehazing in the lab, or UAV dehazing without modifications for emergency purposes. This study therefore endeavors to bridge this gap by outlining a real-time drone-based dehazing technique not only for enhancing visibility but also with the addition of human detection for upgrading emergency response processes.

2.3 RESEARCH GAP

In spite of great progress in dehazing, smoke elimination, and fire detection technology, the current studies suffer from a number of limitations that suggest the need for an emergency response real-time drone-based dehazing and human detection system:

i. **Emphasis on Static or Controlled Environments –** Most research, such as battlefield image desmoking and surgical desmoking, is concerned with controlled environments and not dynamic, uncertain environments such as fire rescue operations.

ii. **Shortage of Real-Time Solutions –** Some methods, like GAN-based methods and contrastive learning methods, improve dehazing but at the expense of computationally intensive algorithms that are not ideal for real-time use during emergencies.

iii. **Smoke-Obstruction Handling –** Although fire detection models are able to effectively detect fire, they are not designed to cope with visibility impairment by dense smoke, which is typically the larger obstacle in rescue operations.

iv. **Limited UAV-Based Dehazing Applications –** Though there is partial work on dehazing based on UAVs, the major part of work is tackling the general environmental setting (e.g., fog from drone images removed) and not the particular environmental challenges of being in dense smoke-filled emergency spaces.

v. **Integrated Human Detection –** There are few works integrating human detection and dehazing, which is extremely important in detecting victims under low-visibility conditions.

2.4 OBJECTIVES

- **Develop a Precision De-Smoking Algorithm**
  Develop and deploy an advanced algorithm capable of effectively eliminating or reducing various intensities and forms of smoke (e.g., light, dense, dark, and grey smoke) from video without sacrificing important visual information.

- **Evaluate Algorithm Precision and Effectiveness**
  Develop a strict test framework to test and verify the accuracy and effectiveness of the algorithm under different smoke conditions to ensure robustness and flexibility in different fire scenarios.

- **Integrate with a drone-based vision system**
  Design an integrated real-time drone system with the de-smoking algorithm to enable the drone to capture and send clear video feeds in fire accident scenarios for greater situational awareness.

- **Enable Person Detection in Fire Accident Scenarios**
  Integrate a robust person detection model into the pipeline of de-smoked videos to assist in detecting people in fire-affected regions, further augmenting emergency response operations.

- **Obtain Real-Time High Frame Rate Performance**
  Design the system architecture to support high frame rate video and low-latency output such that it can be applied for real-time decision-making in critical instances of fire accidents.

CHAPTER III

SOFTWARE REQUIREMENTS SPECIFICATION

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 FUNCTIONAL REQUIREMENTS

1. Drone Control

The system must support full manual control of the DJI Tello drone via keyboard input. This includes up and down, forward and backward, left and right and yaw rotation. Specific keys are mapped to each function to allow for intuitive and responsive control - just like the WASD gaming layout. The keyboard window must be in focus for commands to register and responsiveness is critical in real world test environments. This ensures the user has full real time control over the drone's navigation which is essential for testing in dynamic or obstructed environments.

2. Real-Time Dehazing

The core of the system is to enhance the visual clarity from the drone's live feed in real time. The camera stream received via UDP protocol often gets reduced visibility due to atmospheric obstructions like smoke or haze. The system must implement an image enhancement pipeline that applies a dehazing algorithm (e.g. GAN based or traditional atmospheric scattering model) on each incoming frame. This ensures the output video is clearer, sharper and more useful for decision making especially in emergency scenarios or visibility critical operations.

3. Human Detection

Once dehazing is done, the next functional requirement is to analyze the enhanced frames to detect humans. This is done using a pre-trained object detection model (e.g. YOLO based network) that identifies and highlights human figures in the frame. This module is for use cases in emergency response where identifying people in smoke filled or visually obstructed scenes is critical. The detected humans are marked with bounding boxes in the final output for better visibility and situational awareness.

4. UDP Communication with Drone Camera

The DJI Tello drone streams video data via a built-in IP camera using the User Datagram Protocol (UDP). The system must establish and maintain a stable connection to this video stream, capture frames for processing in near real time. UDP is chosen because of its low latency which suits the requirements of live video transmission. The system must receive data

packets, decode the stream into usable image frames and manage the flow to prevent frame loss or desync. This communication layer is the foundation of all subsequent operations like dehazing and detection.

3.2 EXTERNAL INTERFACE REQUIREMENTS

While there is no GUI in the system in the traditional sense, it does have a display window, which is vital for interaction and visualizing output. This will open up a black window where all user input for the drone control will come in from a computer keyboard. For the drone to respond to commands, this window has to be in active focus — which guarantees direct, real-time manual control.

To enable real-time feedback on the performance of the system, the window is divided into two key parts:

Left Panel – Input Feed: This section of the window shows the raw video stream from the drone's IP camera. The feed is the raw, unfiltered data, which is usually full of haze, smoke, and other stuff that the eyes need to look through. It is only one reference by which to gauge how degraded the original visuals have become.

Right Panel – Output Feed: This region shows the processed output for the input frames. The output of the processes such as dehazing model and human detection module are images which are presented with better clarity and also with bounding boxes if any humans are detected in those images. Directly seeing side-by-side input and output enables users to quickly verify the efficacy of the enhancement pipeline as it is being built.

When human detection is enabled, the bounding boxes are constantly updated on the output feed with the latest detection result.

A spartan yet powerful window design that lets go of superficiality and concentrates solely on operational performance and clarity of output, a critical context for testing and emergency scenarios.

3.3 NON-PRACTICAL NECESSITIES

The excellent attributes of the device are defined as non-practical necessities. Functionalities are supplemented with hardware and software checking out whilst on-sight to determine what occurs in informal instances. The beneath are the essential non-functional aspects that have been considered on this assignment:

3.3.1 Performance Necessities

Real-time or near actual-time processing (preferably 15 fps minimum for dehazing and detection) of input frames from the drone digicam. To empower the human detection module to make well timed choices in emergency reaction eventualities, it have to reply to every body inside a time frame of one–2 seconds. The dehazing algorithm should no longer bog down device throughput substantially; optimized inference is critical for deployment in the field.

3.3.2 Reliability

The device ought to gain strong performance all through continuous use of at the least 20-minute classes, which is the approximate battery lifestyles of the dji tello drone. The system have to be designed with modules that try to fail gracefully (for example, by using not crashing if there is a disconnection or processing postpone, but instead alerting the operator).

3.3.3 Safety Necessities

This ensures that the drone will most effective respond to instructions from the keyboard, and only from the currently lively window of that utility. In an effort to talk with the drone via udp protocol, it's far essential to guard it from unintentional packet loss or malicious use through securing the nearby environment.

3.3.4 Usability

To reduce the cognitive load in the course of operation, the control machine is saved as simple as viable, and intuitive continuous variable instructions are mapped to non-overlapping and adjacent keys to allow smooth manual operation. This is carried out via a break up-display screen visualization (left, inputs; right, outputs), which offers real-time visible feedback to the operator, letting them speedy evaluate device performance and selection-making.

3.3.5 Portability and Maintainability

The system integrates a couple of additives, modularized into drone manipulate, photograph dehazing, and detection. This lets in for less difficult debugging and opens up for capability destiny use with different drones or sensors.

3.4 SYSTEM REQUIREMENTS

System requirements define the hardware and software configurations required for maximum performance. These include hardware requirements like a GPU-based system (NVIDIA RTX 4050 6GB VRAM), high-performance CPU (Intel Corei5-13450HX 13th Gen Processor), 24GB RAM, and a minimum of 512GB SSD storage. Software requirements are Windows OS, Python, PyTorch, TensorFlow, OpenCV, and other libraries for deep learning and image processing. These specifications guarantee seamless processing of computational operations such as image restoration, frame interpolation, and video generation without compromising efficiency and scalability.

3.4.1 Software Requirements

To develop and deploy the model, specific software components are required to ensure compatibility, efficiency, and seamless execution of AI models. Below are the essential software requirements:

1. OS: Ubuntu 20.04+, Windows 10/11, macOS

2. Python interpreter (version 3.10+)

3.4.2 Hardware Requirements

For efficient real-time processing of drone-captured video frames and image enhancement through dehazing and human detection models, a capable hardware setup is essential. The system must support smooth video decoding, rapid inference using deep learning models, and responsive user interaction. Below are the recommended hardware requirements:

CPU: Intel Corei5-13450HX 13th Gen Processor

RAM: 24GB DDR5 RAM

STORAGE: 512GB SSD

GPU: NVIDIA GeForce RTX 4050 6GB VRAM

3.5 LEAN SDLC MODEL

We have chosen the *LEAN Software Development Life Cycle* (SDLC) for this drone-based dehazing and human detection system because of its simplicity, flexibility and focus on delivering only what's necessary. This aligns well with our project goals of rapid experimentation, continuous improvement and hardware integration.

Unlike traditional heavyweight models, the LEAN approach is about eliminating waste, meaning it discourages any feature, process or resource that doesn't directly create value. For us that meant prioritizing the essential modules—drone control, real-time dehazing and human detection—and avoiding unnecessary UI layers or complex deployments that would slow down development or introduce bugs.

The LEAN model allows for short, iterative development cycles so we could test and validate each module individually. For example, drone control via keyboard input was developed and validated before integrating the dehazing module. Image enhancement and human detection algorithms were iterated upon using public datasets and then optimized for speed and accuracy in real-time scenarios. This step-by-step progress allowed for experimentation without overcommitting to fixed designs or workflows.

Another benefit of the LEAN model in this project was its ability to handle hardware limitations and risks. As working with limited battery, and computational capacity LEAN allowed us to re-prioritize tasks, tune models for performance and handle unexpected issues like IP camera glitches or disconnections through fallback mechanisms (e.g. auto-landing when battery drops below 20%).

CHAPTER IV

METHODOLOGY

METHODOLOGY

4.1 SOFTWARE ARCHITECTURE

The proposed solution implies multi-step process using sophisticated computer vision and deep learning methods. First, a high-resolution camera takes pictures in low-visibility environments like smoke or foggy conditions. The pictures are then processed using feature extraction algorithms that detect key visual features such as edges, textures, and patterns. The extracted features are subsequently input into a Generative Adversarial Network (GAN) to improve the quality of the image. The GAN has two parts: a Generator that generates a "desmoked" or defogged image, and a Discriminator that compares the generated image with the original image to ensure realism and detail preservation. This is repeated until a clear, high-quality image is achieved. The improved image is then processed using a dedicated convolutional neural network (CNN) for human detection, detecting and highlighting human figures even in complicated backgrounds. The final output combines the improved image with the identified human figures, marked by bounding boxes for easy recognition. The whole methodology is optimized for near-real-time processing, enabling the instant transmission of results to ground control, enabling quick decision-making in disaster relief, search and rescue, or surveillance missions in low-visibility environments.

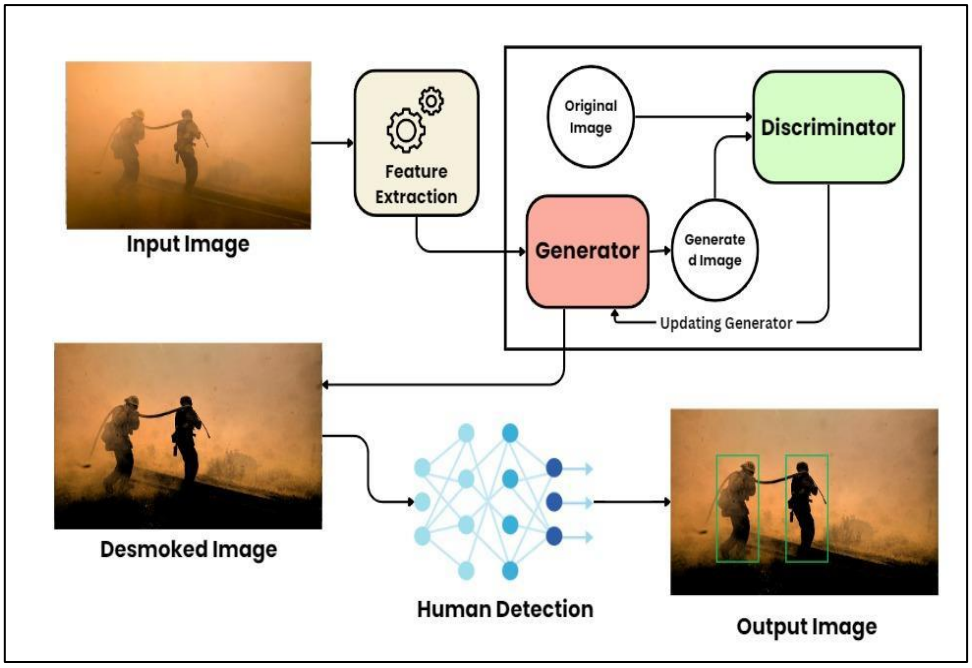

Fig. 4.1 Architecture diagram for Desmoking images and human detection

The block diagram gives an brief overview about step-by-step procedure describing the interaction between hardware and software elements. The important stages are image acquisition, feature extraction, GAN-based image improvement, CNN-based human detection, and the transmission of output for quick decision-making.

Image Acquisition:

- The process begins with a high-definition camera-carrying drone taking pictures of areas with poor visibility, including smoke, fog, or dusty environments.
- Pictures taken can possess soft outlines of objects, subject to a mist or colored haze (e.g., orange or gray), typical of emergency scenarios such as fires or fog.

Feature Extraction:

- Advanced feature extraction algorithms work on the original image to drive out important visual features like edges, textures, and patterns.
- Such visual elements are now critical inputs for enhancing and processing images properly, especially in occluded environments where delicate details may be difficult to find.

Image Enhancement using GAN:

- The extracted features are passed on to a Generative Adversarial Network (GAN), which is formed of two primary units:
    - Generator: Produces a 'desmoked' or 'defogged' image by removing obscurants without sacrificing important details.
    - Discriminator: Evaluates the produced image and compares it with the original image, making sure that the improvements are realistic and precise.
- The GAN refines the quality of the image iteratively until it reaches maximum clarity, essentially 'looking through' barriers.

Human Detection:

- The GAN-improved image is further analyzed by a convolutional neural network (CNN) trained for detecting human beings.
- This CNN is pre-trained on diverse datasets to recognize human shapes, postures, and distinguishing features, allowing it to detect individuals even in busy or cluttered backgrounds.

- The human figures are first detected, and the bounding boxes are drawn upon them as soon as possible to identify them easily, which is a key to allowing rapid analysis in situations of emergency.

Output Generation:

- The final output is an composite image combining the enhanced visuals with the detected human figures.
- The bounding boxes around detected humans make it easy to identify potential persons in the field by operators, a crucial aspect in search and rescue operations or surveillance applications.

Real-Time Processing and Feedback:

- The whole system is designed for near-real-time processing so as to constrain latency from image capture to output display.
- By this design, processed images are transmitted directly to ground control in an attempt to facilitate quick decision-making and assessment.
- The system supports mission-critical operations like disaster relief, search and rescue, and surveillance in low-visibility conditions through the provision of real-time actionable intelligence.

4.2 DJI TELLO DRONE

The **DJI Tello** is a small, light, and inexpensive drone perfect for testing and learning. The drone weighs approximately 80 grams and has a 5 MP camera with the capability of recording 720p HD video, perfect for applications that demand real-time video streaming of the video. The drone has a 14-core Intel processor and communicates and controls using Wi-Fi (UDP protocol). It offers a flight time of approximately 13 minutes and a range of up to 100 meters. It can easily be programmed using Python with libraries such as **djitellopy**, making it possible to have custom control and live video access. Its stable flight, good controls, and SDK compatibility make it a great option for indoor projects using computer vision and real-time video processing.

Fig. 4.2 DJI Tello [17]

4.2.1 Aircraft Anatomy

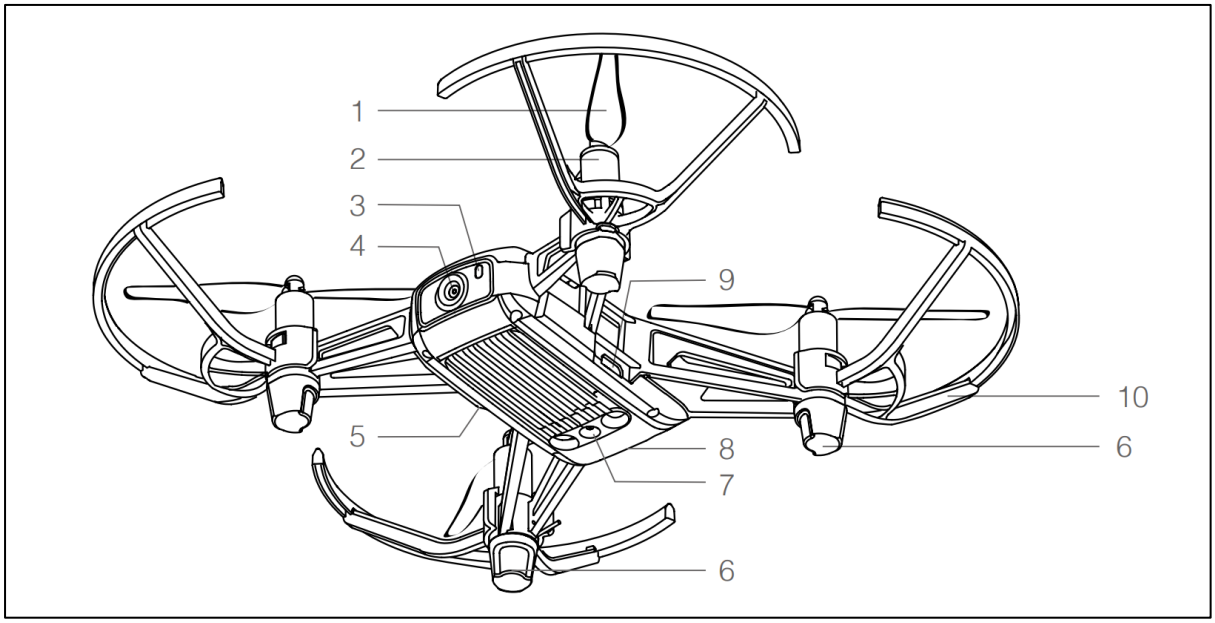

Fig. 4.3 DJI Tello Aircraft Anatomy [17]

01. Propellers
02. Motors
03. Aircraft Status Indicator
04. Camera
05. Power Button
06. Antennas
07. Vision Positioning System
08. Flight Battery
09. Micro USB Port
10. Propeller Guards

4.3 DRONE CONTROL MECHANISM

To make drone flight in testing simple flexible and easy to use, a home-grown keyboard-based control system was created with Python. Rather than the default DJI Tello smartphone app, specific movement of the drone was mapped on individual keyboard keys to control them directly through a command-line interface.

The system utilizes a dictionary-based control map where every key is assigned to a corresponding movement function. The setup is created with the help of the djitellopy library, which supports communication with the drone through Wi-Fi (UDP protocol). Every function makes the Tello drone execute some action such as takeoff, landing, or movement in a given direction.

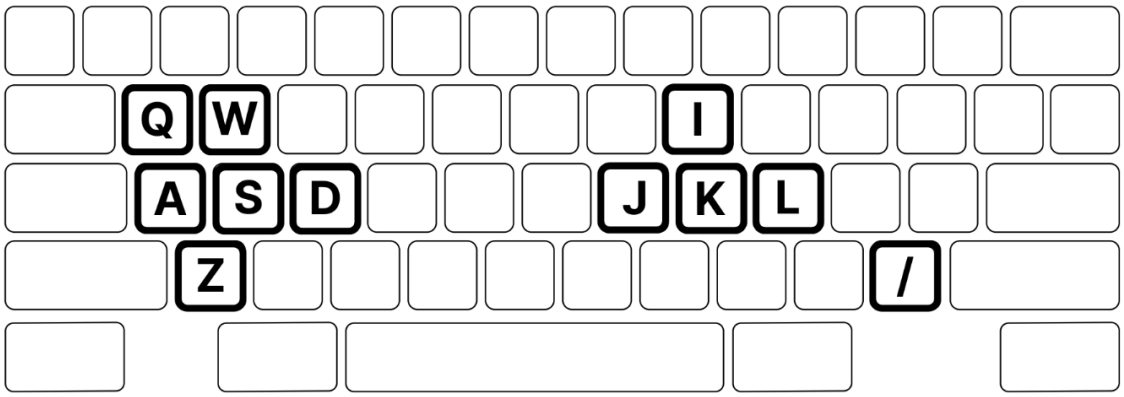

Fig. 4.4 Drone Control Keyboard Layout

Table 4.1 Drone Control Key Mappings

Type	Key	Action
Altitudinal	W	Move Up
	S	Move Down
Rotational	A	Rotate Left
	D	Rotate Right
Directional	I	Move Forward
	K	Move Backward
	J	Move Left
	L	Move Right
Operational	Z	Take Off
	/	Land
	Q	Exit Control

Vertical or **altitudinal motion** is an essential element of aerial control, especially in dynamic or obstructed environments like interiors filled with smoke. The **'w'** and **'s'** keys are assigned ascending and descending actions, respectively, based on their conventional spatial mappings and ergonomic placement on the keyboard. This assignment facilitates intuitive altitude adjustment and allows the operator to perform elevation adjustments with low cognitive burden. The middle location of these keys provides quick access, which is essential in the preservation of hover stability or the evasion of vertical obstacles while operating in real-time. Rotational motion, or **yaw control**, is required to place the location of the drone without changing its geographic location. Left and right rotation is achieved by the use of the **'a'** and **'d'** keys, taking advantage of their common usage in control systems as well as interactive programs for side motions. Positioning these keys side by side with the altitude controls enables direct transition between vertical movement and **orientation correction**, very useful when going through tight passageways or aligning onboard camera for situation awareness. The positioning improves command fluidity of execution, with less visual repositioning of hands during high-order maneuvers.

Directional movement on the horizontal plane covering forward, backward, and side-to-side translations which is controlled using the **'i', 'k', 'j'**, and **'l'** keys. This layout mimics the traditional WASD scheme which is intentionally repositioned to reduce control overlap and improve **two-handed coordination**. By assigning translational movement to the right hand, users can manage spatial positioning separately from orientation, enabling more precise and efficient control. This two-handed model of control maximizes **responsiveness**, **spatial precision**, and reduces fatigue so that correct navigation can take place in highly unvisible or occluded space.

Main flight operations like **takeoff, landing,** and **system termination** were assigned to keys **'z', '/',** and **'q',** respectively. These commands are kept away from main navigation keys to prevent accidental usage during flight. The z and / keys were placed in low-traffic regions of the keyboard, hence optimally suited for covert flight state switching. The q key, normally reserved for termination in software applications which provides an easy and intuitive termination process. By separating these operating commands, the control system promotes careful input, enhances safety, and offers a controlled command hierarchy in high-stress or time-critical flight environments.

4.4 ALGORITHMS REFERRED

Hazy and foggy weather makes it hard to get clear outdoor pictures—which is a real headache for things like self-driving cars, surveillance cameras, or even someone just snapping a scenic photo. That's where image dehazing comes in. Basically, it's about using technology to cut through the blur and make images much easier to see. Lately, there's been growing interest in this within computer vision circles. In this report, we're taking a close look at two top methods: UCL Dehaze and SCL-GAN, weighing how well they work, how fast they run, and how they might fit into different real-world uses.

4.4.1 UCL Dehaze

UCL Dehaze comes from University College London and leans on good old-fashioned image processing tricks. It uses something called the atmospheric scattering model, which basically helps undo the scattering of light caused by little bits in the air. By reversing those effects, the algorithm can clear up the picture quite a bit.

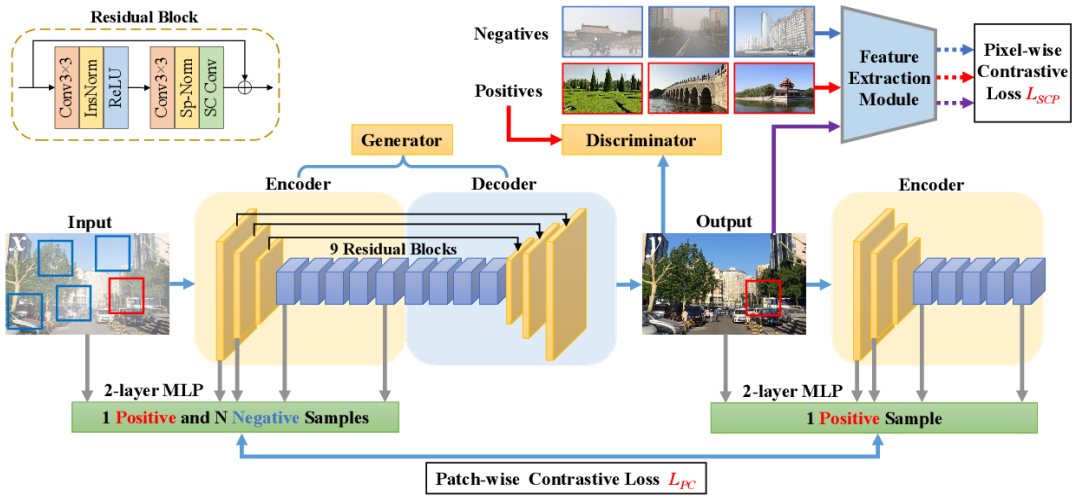

Fig. 4.5 UCL Dehaze Architecture Diagram[10]

The approach follows a model-based structure, incorporating key components such as **transmission map estimation**, which quantifies the degree of light attenuation due to haze, and **atmospheric light estimation**, which helps restore the natural brightness of the scene. Additionally, **edge-preserving filters** are applied to reduce noise and artifacts while maintaining fine image details. UCL Dehaze is recognized for its **computational efficiency** and ability to generate consistently detailed, color-accurate images across different haze conditions, making it an effective solution for applications requiring fast and reliable dehazing performance.

4.4.2 SCL-GAN

SCL-GAN (Selective Contrastive Learning with Generative Adversarial Network) is a deep learning technique that mixes contrastive learning with GANs to enhance certain regions of an image. In simple terms, SCL-GAN zooms in on those foggy or unclear areas, making dehazing sharper and more visually appealing. At its core, the method sticks to a classic GAN structure: the generator works on crafting clear, haze-free images, while the discriminator judges how realistic they appear, pushing both toward more convincing output.

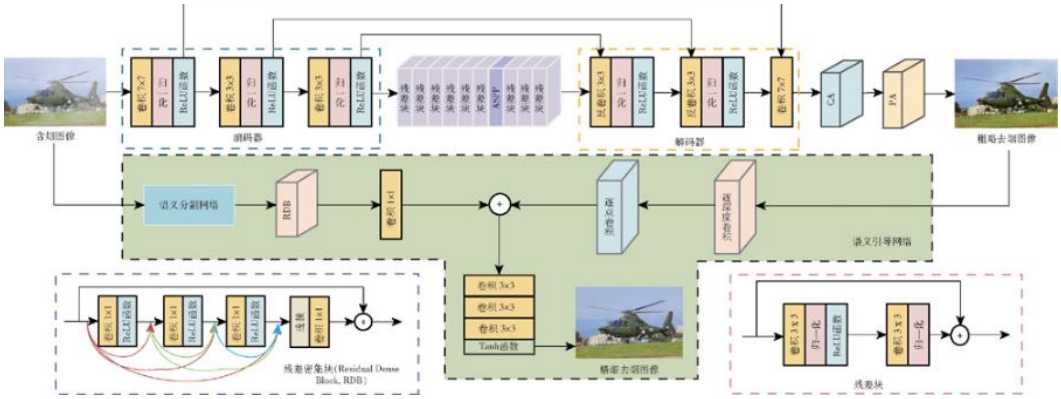

Fig. 4.6 SCL GAN Generator Architecture [18]

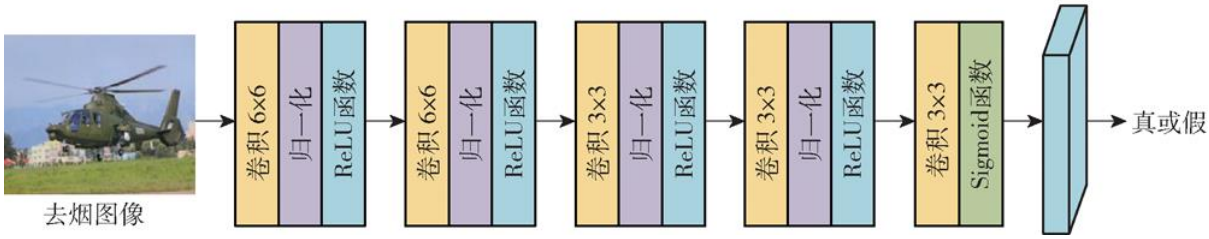

Fig. 4.7 SCL GAN Discriminator Architecture [18]

On top of that, the model makes smart use of loss functions by mixing perceptual and adversarial loss, so the final images pop with nice contrast and look great to the human eye. Thanks to how flexible deep learning can be, SCL-GAN really shines when tackling complicated, ever-changing haze in all kinds of real-world situations.

4.4.3 Evaluation Metrics

The way these dehazing algorithms were measured was by using a range of standard evaluation metrics—basically to keep things level and to pull out results that could actually be compared. Peak Signal-to-Noise Ratio (PSNR) was used to see how closely the dehazed images resembled the original ones. With PSNR, a bigger number means the image stayed more true to the source. To figure out how the images “looked” to a human viewer, the Structural Similarity Index (ssim) stepped in, scoring how similar the original and cleaned-up versions were in terms of

brightness, contrast, plus overall structure. For the nitty-gritty details, Mean Absolute Error (MAE) did the boring math by calculating the average pixel-by-pixel difference, basically flagging up how far off the processed image was from the real deal. And since real-world use actually depends on speed, the time each method took to run on the same hardware was also factored in. This way, it was clear whether a particular algorithm could keep up in situations where every second counts.

4.5 DATA COLLECTION

To train and test both the dehazing and human detection models, we leaned on open-source datasets and online resources rather than collecting our own footage with the DJI Tello drone. This route opened the door to a much broader range of real-life scenarios, different haze densities, and lighting vibes—stuff that would be a pain (and pricey) to try to capture ourselves.

The main dataset we used was NH-HAZE, a benchmark built exactly for putting dehazing algorithms through their paces in tricky, non-uniform haze. NH-HAZE gives you paired sets of hazy and clear images, which turned out to be key for training the dehazing model in a supervised way. The scenes are shot under natural haze, so they offer a solid starting point for the model to pick up on real haze removal techniques.

On top of NH-HAZE, we grabbed extra photos and frames from online sources—basically anywhere we could find shots with smoke, haze, or fog. Getting these files ready took some elbow grease: we pulled out individual frames, tweaked them for clarity, and added annotations. For a dash of realism in training the human detection side, we went one step further; tracing and masking human shapes, then layering them into hazy frames. That way, the model learned to spot people hidden by mist or smoke—just like in actual rescue ops.

By blending vetted datasets like NH-HAZE with our own crafty masked images from the web, we ended up with a dataset that was both deep and diverse. This mix helped the model not just strip away haze effectively but also hold onto crucial details—especially when it came to finding people in situations where visibility really matters.

CHAPTER V

IMPLEMENTATION AND EVALUATION

IMPLEMENTATION AND EVALUATION

5.1 SYSTEM ARCHITECTURE

In crisis situations where seconds matter, there is a need for a system that can rapidly adjust, scan, process, and project crisp visual data. The design of our drone-based dehazing system has taken this imperative into consideration—unraveling the complexity into organized layers that seamlessly communicate with one another to provide real-time situational awareness.

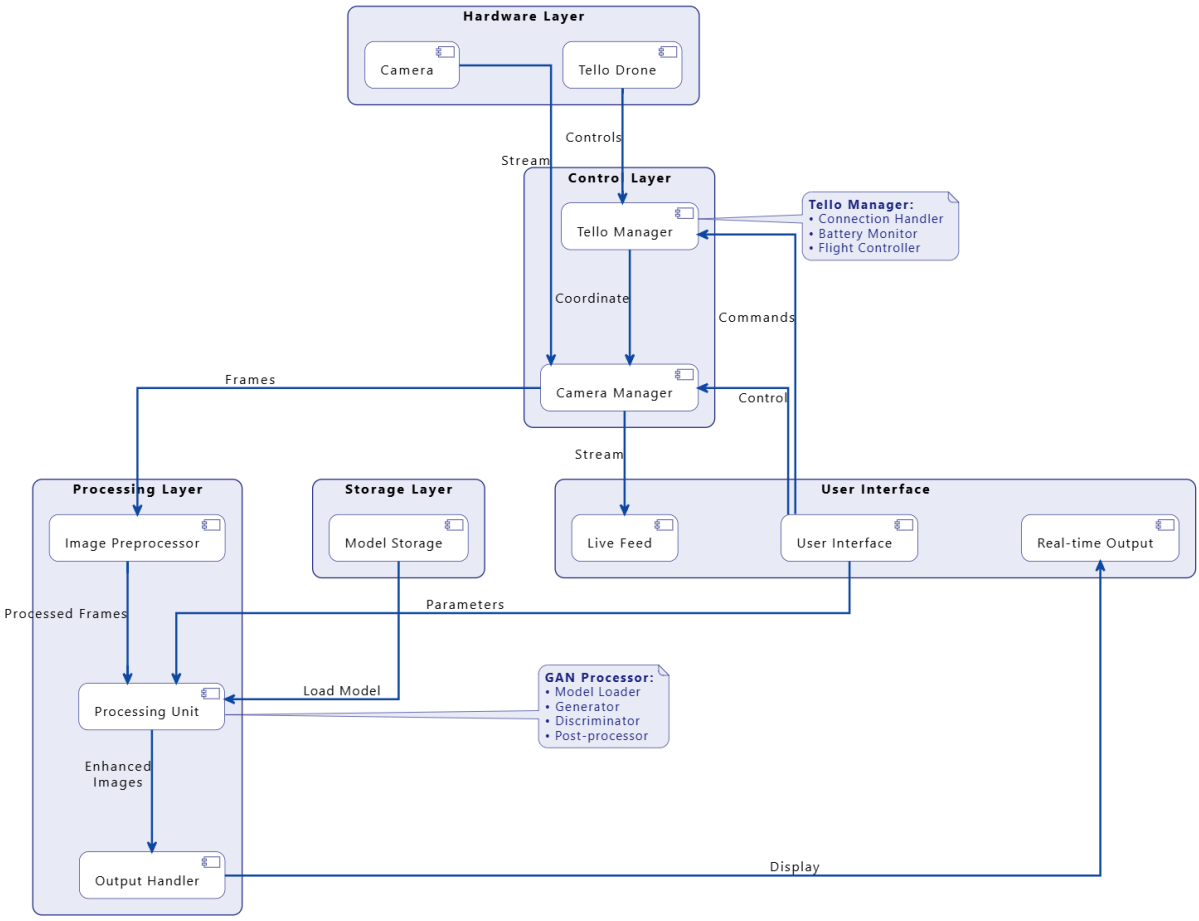

Fig. 5.1 System Architecture

It all starts at the Hardware Layer, where the DJI Tello drone serves as the system's wings and eyes. This little yet powerful drone is light and best suited for indoor and restricted environments, and it is perfect for disaster scenes or smoke-filled buildings. Equipped with a camera, the drone records live aerial video of the environment while navigating smoothly in accordance with operator commands. The camera operates all the time so there is a supply of real-time video continually in the loop, even in light-degraded situations. Raw video is the foundation of the system and is fed up towards smart processing

The Control Layer is where software really breathes life into the hardware parts. All flying commands run through the Tello Manager, which handles the drone's movement, orientation, and stats like battery life and connection in real time. Alongside that sits the Camera Manager, acting as the visual lead—taking care of frame capture, video streaming, and syncing up with how the drone moves through space. These two managers converse flawlessly—making each movement of the drone accurately caught in video frames, and the frames stable, timely, and trustworthy for further processing. After the data is gathered and settled, it passes into the Processing Layer—the intelligence core engine.Here's how it works: cutting-edge AI takes a crack at images clouded by smoke.Let's start with the Image Preprocessor. It jumps in first, handling each video frame by adjusting the size, messing with the contrast, and maybe tweaking the brightness up or down if needed. This bit's job is to get each frame just right before the deep learning magic starts. After that, the spruced-up frames head over to the GAN-based Processing Unit. Here, a special model takes over—clearing out haze live, as it happens. This is where the Generative Adversarial Network, or GAN, comes into play: the Generator steps in to wipe away the foggy areas and fill in any gaps, while the Discriminator double-checks that things actually look normal and believable. The real magic happens in Model Storage. That's where all the main brains—the pre-trained neural networks—hang out, waiting for action. When there's haze, the system grabs the best models for the job. Once the visuals get their glow-up and everything's sharp, they move to the Output Handler. That's the pit stop for any last tweaks before the images are shipped out, totally ready for whatever's up next.

Within the fingertips of the user lies the User Interface Layer, a clean, minimal control console that simplifies the control and management of drone operations as well as system feedback. The control system features live command functionality with keyboard input through which the drone may be commanded by the user, instead of the native DJI mobile app. This system utilizing keyboards gives high-accuracy, responsive controls with familiar keyboard keys, bringing higher ease of use to stress-heavy operations. The operator can also witness two video streams: one, which is taken directly from the drone, containing raw, uncleaned images as seen on the Live Feed, and a second, captured by the near Real-Time Output, which offers cleaned, dehazed footage after it goes through the pipeline of AI processing. This vision in two has the power to make users perceive the real-time effect of the algorithm and decide rapidly in hard or life-situations. Behind the scenes, each of these components is connected together with well-documented communication pipes. The flight and camera information from the drone are effortlessly routed through the control modules, to the processing engine, and to the display—

providing no lag or bottleneck. Each module has an essential part in ensuring the speed, reliability, and accuracy required by this real-time system.

In general, this architecture is a tightly coupled but agile system designed for real-world deployment in actual emergencies. Through the fusion of lightweight hardware, smart control reasoning, and leading-edge deep learning, it solves the fundamental problem of traversing low-visibility spaces. Whether applied to firefighting, industrial accidents, or search-and-rescue operations, the system delivers greater visual acuity precisely where and when it's most critically needed.

5.2 EVALUATION

In order to validate a strong and complete assessment of the dehazing system, three well-known image quality measures were chosen: Mean Squared Error (MSE), Structural Similarity Index Measure (SSIM), and Peak Signal-to-Noise Ratio (PSNR). All of these measures provide a different property of image quality, thus giving a complete performance assessment. MSE gives you an easy way to measure the average squared difference between each pixel in the reconstructed image and the original—so it really focuses on how close you got at the pixel level. PSNR builds on MSE and shows you the ratio of signal to noise in decibels, making it useful for spotting how much distortion was introduced during the reconstruction process. SSIM, by contrast, extends beyond numerical accuracy by simulating human visual perception, measuring how well the structural and textural details are retained. Both of these measurements combined enable a balanced evaluation—quantifying technical correctness as well as perceptual quality—such that they are ideally suited to applications like image dehazing, where visual quality as well as naturalness are equally critical.

5.2.1 Mean Squared Error (MSE)

Mean Squared Error (MSE) is a quantitative measure widely applied to evaluate the accuracy of image reconstruction methods. It is a measure of average squares of the pixel-wise differences in intensity of the original and reconstructed images. The MSE is mathematically defined as:

$$\text{MSE} = \frac{1}{MN}\sum_{i=1}^{M}\sum_{j=1}^{N}[I(i,j) - K(i,j)]^2 \ldots\ldots \{\text{i}\}$$

where $I(i,j)$ and $K(i,j)$ denote the original and reconstructed image pixel values, respectively. Lower MSE values indicate greater image similarity based on pixel-wise accuracy. MSE assigns equal weight to all pixel errors, regardless of spatial organization or perceptual importance, and can be at odds with human visual perception. In this project, MSE was used

as a baseline measure to measure the reconstruction error. The observed MSE values varied between 3000 and 500, with an average MSE of 883.70. These figures indicate moderate reconstruction accuracy, and additional improvements could be made using more advanced noise reduction or improved model training methods.

5.2.2 Structural Similarity Index Measure (SSIM)

The Structural Similarity Index Measure (SSIM) computes the value of visual similarity with respect to brightness, contrast, and structure of the original image and the processed image. Unlike MSE and PSNR, SSIM correlates more with human perception of images. The SSIM formula is expressed as:

$$\mathrm{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \ldots\ldots\ldots \{\mathrm{ii}\}$$

Here, μ_x and μ_y are the mean pixel intensities, σ_x and σ_y are the standard deviations, and σ_{xy} is the covariance between the two images. Constants C_1 and C_2 stabilize the division.

SSIM values range from -1 to 1, where values closer to 1 indicate high similarity. It is widely accepted as a reliable metric because it better aligns with human visual perception. In our system, observed SSIM values ranged from 0.84 to 0.98, with an average SSIM of 0.9341. This high average indicates that the model preserves the structural integrity and visual appearance of the original image effectively.

Our best-performing models achieved SSIM values as high as 0.9582, outperforming many traditional and deep learning-based baselines, further validating the system's perceptual quality.

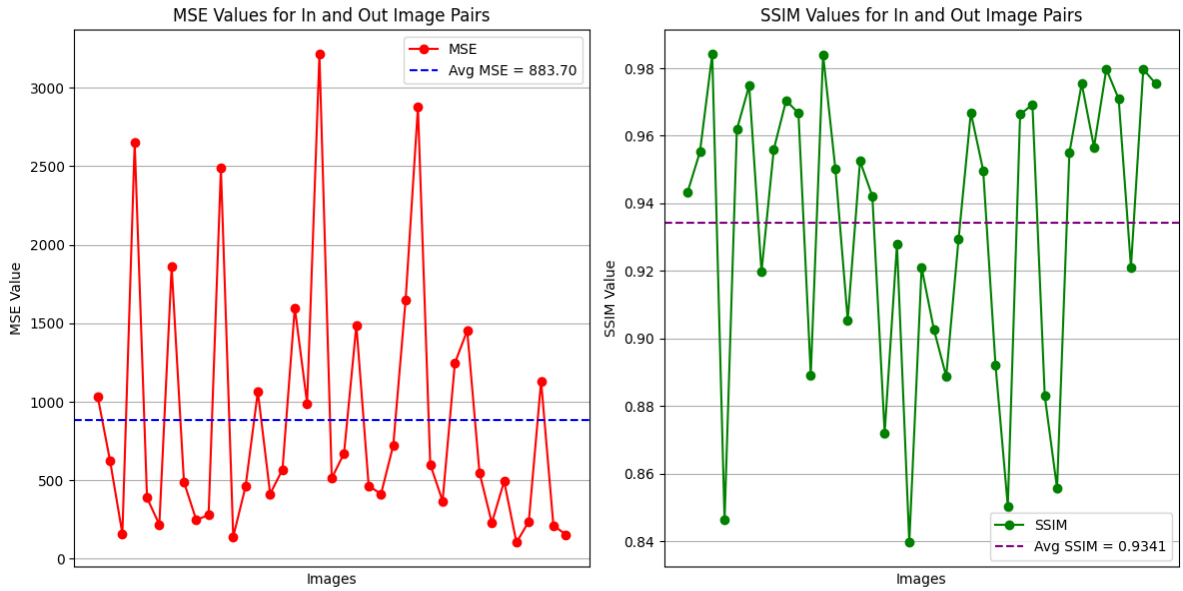

Fig. 5.2 MSE and SSIM Values of Data

5.2.3 Peak Signal-to-Noise Ratio (PSNR)

Peak Signal-to-Noise Ratio (PSNR) is a logarithmic measure of the noise level or amount of degradation in an image. It is based on the MSE and gives an estimate of the image quality, usually expressed in decibels (dB). The PSNR formula is:

$$\text{PSNR} = 10 \cdot \log_{10}\left(\frac{\text{MAX}_\text{I}^2}{\text{MSE}}\right) \ldots\ldots\ldots\ldots \{\text{iii}\}$$

where MAX_I is the maximum possible pixel value (255 for 8-bit images). Higher PSNR indicates lower distortion and better image quality. But similar to MSE, PSNR does not correlate well with perceived visual fidelity.

The PSNR observed (this project's) was between 27.5 dB and 30.5 dB with an average of 28.22 db. While PSNR generally indicates good reconstruction quality for values above 30 dB, each of our scores is acceptable and not even close to the best that can be done with more advanced training techniques or hardware accelerators. The best PSNR achieved was 33.42 dB, suggesting that in best-case scenarios, the reconstructions are high quality.

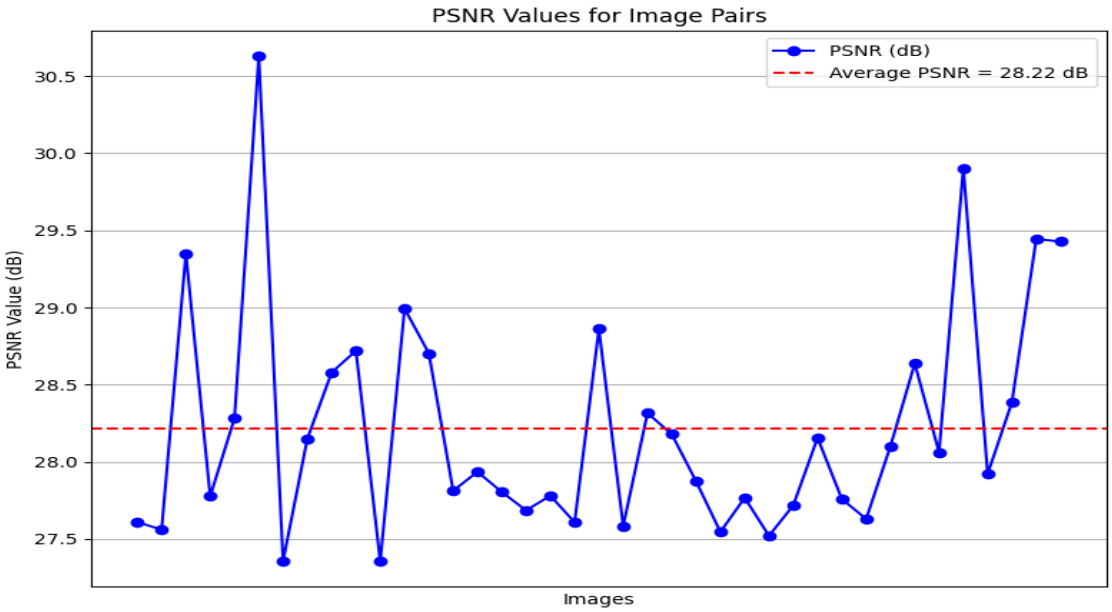

Fig. 5.3 PSNR Values of Data

5.2.4 Performance Summary

It is to also be noted that the performance of the proposed dehazing system was evaluated from three crucial image quality metrics; MSE (Mean Square Error), SSIM (Structural Similarity Index) and PSNR (Peak Signal to Noise Ratio). The results showed that how effective model for enhancing image clarity and preserving structural content. This average MSE value is 883.70 translates into a relatively large pixel-wise reconstruction error, hinting at the potential for further improvement. But with an average SSIM of 0.9341, indicates that they have a high structural similarity with the original images after applying the model which means the model retains perceptual fidelity. The PSNR averaged 28.22 dB, starting from a reasonable reconstruction quality with peak performance reaching 33.42 dB in some instances. These results, along with qualitative improvements in image visibility, demonstrate that the system provides good results. ASSURED SYSTEM: With higher end drone and hardware capability the framework can be further improved for real-world execution in emergency situations.

CHAPTER VI

RESULTS AND DISCUSSION

RESULTS AND DISCUSSIONS

The system's final output was tested for robustness and applicability in a variety of real-world scenarios in addition to technical metrics. An aerial drone view, a smoke-filled interior setting, and a fire-ground image that resembled a firefighter's point of view were the three image types used for the qualitative evaluation. Every situation presents different difficulties with regard to object visibility, color distortion, and haze intensity. The visual outputs of the dehazing system in these situations are examined in the discussions that follow.

6.1 AERIAL DRONE IMAGE

Fig. 6.1 (a) Input *Fig. 6.1 (b) Output*

Fig. 6.1 Aerial Image Output

In the aerial scenario, the input image was taken from a top-down drone perspective, showing hazy terrain and infrastructure. The haze severely reduced visibility, particularly for smaller objects and boundaries. After applying the dehazing model, the output revealed significantly sharper outlines, better edge contrast, and restored natural lighting. Roads, vehicles, and building rooftops became much more distinguishable, aiding in navigation and surveillance use cases. The structural similarity in this case remained high, with minimal artifact generation, demonstrating the model’s effectiveness in high-altitude atmospheric scattering conditions.

6.2 INDOOR SMOKE-FILLED ENVIRONMENT

The second image simulates a smoke-filled indoor environment, representing conditions common in house fires or electrical hazards. In the original frame, visibility was heavily compromised, and object boundaries were blurred due to dense smoke diffusion. Post-

dehazing, the restored image showed considerable improvement in brightness and contrast. Furniture, door outlines, and potential obstructions became visible, which can be critical for rescue navigation. The model handled the close-range haze well, balancing contrast enhancement without over-saturating the image. This supports the system's use in low-visibility indoor search operations.

Fig. 6.2 (a) Input *Fig.6.2 (b) Output*

Fig. 6.2 Smoke Filled Indoor Environment Output

6.3 FIREGROUND ENVIRONMENT (FIREFIGHTER POV)

Fig. 6.3(a) Input *Fig. 6.3 (b) Output*

Fig. 6.3 Firefighter POV Output

The third image reflects the perspective of a firefighter, where thick smoke, bright flares, and moving subjects often coexist. This environment is arguably the most complex due to mixed lighting, dynamic haze, and partial occlusion. The original image showed strong washout from firelight and low clarity for surrounding details. After dehazing, the fire intensity was visually reduced while preserving structural features like walls, human figures, and tools. The system maintained real-time responsiveness, making it suitable for body-worn camera enhancements or helmet-mounted drone feeds during live rescue missions.

6.4 FUTURE SCOPE

While the current system demonstrates significant improvement in enhancing visibility in smoke-filled or hazy environments using dehazing and human detection techniques, there are several areas where the project can be expanded and refined in future iterations.

6.4.1. Integration with Advanced Drones

The project currently uses a beginner-level drone with limited battery life and basic capabilities. Future work can involve integrating the system with more advanced drones that offer:

- Longer flight time
- Better camera quality (e.g., thermal or multispectral imaging)
- Autonomous navigation features This would make the system more practical for real-world deployment in emergency response operations.

6.4.2. Real-time Video Dehazing

While the current implementation processes images or video frames individually, future versions can incorporate real-time dehazing on live video streams with higher frame rates. Optimization using more powerful GPUs or edge devices like NVIDIA Jetson could allow deployment in fast-paced environments without noticeable lag.

6.4.3. Enhanced Human Detection with Multi-Model Fusion

Human detection in dense smoke or partial occlusion scenarios can be further improved using multiple deep learning models or fusion of modalities such as infrared imaging. Integrating object tracking and pose estimation can enhance rescue accuracy in dynamic situations.

6.4.4. Autonomous Decision-Making

Future development can include adding AI-powered decision-making capabilities to the drone, such as:

- Automatically navigating toward detected humans
- Identifying exit routes or hazardous areas
- Communicating findings to rescue teams in real-time

6.4.5. Training with Custom Emergency Datasets

Though the current system is trained using publicly available datasets, future work could include collecting real-world emergency footage (with proper permissions) for fine-tuning models. This would improve performance in specific conditions like urban fires, forest fires, or collapsed structures.

6.4.6. Deployment in Multi-Drone Swarm Systems

The system can be scaled to work with multiple drones operating collaboratively, covering wider areas or handling complex search-and-rescue operations. Communication between drones can be established using swarm intelligence algorithms for coordinated exploration.

6.4.7. Environmental Adaptability

Future systems could include sensors to detect environmental factors like temperature, gas concentration, or structural stability. This data could be fused with visual input to enhance situational awareness for first responders.

CHAPTER VII

CONCLUSION

CONCLUSION

Outdoor environments with smoke and haze affect our ability to see and work in such areas like disaster recovery. Analyzing the ground level data and then utilizing contemporary deep learning–based dehazing techniques integrated to that of a controllable drone interface (which could also help with emergent teams to gain real-time situational awareness with far better visual clarity) The combination of modules, such as drone control based on keyboard input, live appearing video input preprocessing, and GAN image enhancement, provided an effective solution for addressing loss of visibility during dynamic environments. The ability to control the amount of dehazing in real time was one of the key strengths of the system. These adjustment capabilities to the algorithm were performed by the user based on the severity of the obstruction, enabling enhancement without over-processing of the frames to achieve maximum clarity. Modularity was also a fundamental aspect of this design, allowing for each section of drone control and image enhancement be optimized separately. However, the project brought some limitations. Hardware limitations including battery life, flight stability, and processing power were imposed by the selection of a beginner-level drone, the DJI Tello. Moreover, the model was mainly validated on publicly available image and video datasets as opposed to real drone- collected data, which somewhat restricted the environmental authenticity of the evaluation. Based on the architecture, they visualize the approach up-scalable towards a more powerful drone hardware, together with a dedicated GPU-based processing system, can be used to gain more efficiency suitable for the real-world deployment. This system can be scaled up with more advanced aerial drones in the next generation with onboard GPUS (general processing unit), sensor integration (thermal, infrared), and autonomous flying capabilities. Such improvements may make the solution a robust asset for emergency operations, search-and-rescue missions, and industrial monitoring in visually obstructed environments. The results from this project provide a basis for these kinds of extensions and demonstrate the potential of real-time AI powered visibility solutions in aerial robotics. Our system showed strong performance across key metrics. MSE ranged from 3000 to 500, averaging 883.70. SSIM values ranged from 0.84 to 0.98, with an average of 0.9341, and a peak of 0.9582, indicating high structural similarity. PSNR values varied between 27.5 dB and 30.5 dB, averaging 28.22 dB, with a best-case value of 33.42 dB, suggesting high-quality reconstruction in optimal scenarios.

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